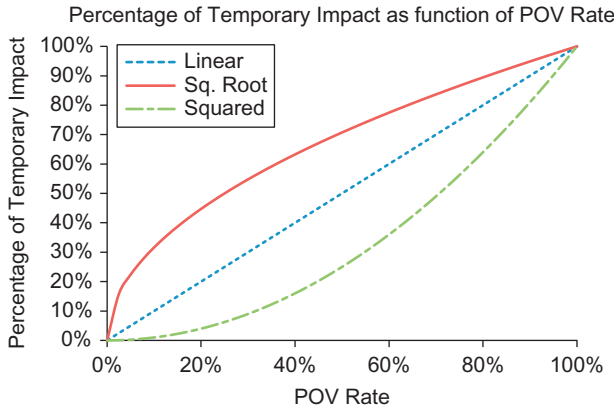




■ **Figure 4.5** Percentage of Temporary Impact as a function of POV Rate.

change at the same rate. When $a_4 = \frac{1}{2}$ the relationship is a squareroot function. Costs increase much quicker for the lower POV rates and at a reduced rate for the higher POV rates. When $a_4 = 2$ the relationship is a squared function. Costs increase at a faster rate for the lower POV rates and at a slower rate for the higher POV rates. Depending upon the rate of change, the a_4 parameter, investors may structure their underlying trading schedule in different ways, either trading faster or slower than normally. Notice that at $\text{POV} = 100\%$ investors incur the entire temporary cost regardless of the function of the temporary impact rate.

Up to this point we have not yet defined the functional form of I^* . Our practical experience, empirical evidence, and data observations have found that trading cost is dependent upon size, volatility, and strategy. Thus, our functional form needs to include at least these variables. Some alternative or competing models have included market cap or other stock fundamental factors to differentiate between costs for different stocks even for the same relative size, e.g., 5% ADV. Our analysis has found that volatility provides a better fit than variables such as market cap, log of market cap, etc. At the very least, we need to ensure that the model adheres to the essential properties defined above.

***I** Formulation**

Our preferred functional form for I^* is the following power function:

$$I_{bp}^* = a_1 \cdot \left(\frac{Q}{ADV} \right)^{a_2} \cdot \sigma^{a_3} \quad (4.13)$$